

RITESH JANGA

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RESEARCH PROFILE

Computer Science PhD researcher developing **reliable, efficient, and auditable AI systems** at the intersection of multi-agent LLMs, retrieval, model evaluation, and privacy-preserving machine learning. Builds research prototypes and reproducible benchmarks in **Python, PyTorch, TensorFlow, RAG/GraphRAG, federated learning, and statistical evaluation**. Author or co-author of six publications with experience translating research questions into evaluated systems.

EDUCATION

University of Cincinnati – PhD in Computer Science (Data Science)	Jun. 2026 – May 2030
Minnesota State University, Mankato – MS in Data Science, Thesis	Aug. 2024 – May 2026
Vignan’s Institute of Information Technology – BTech in AI and Data Science	Aug. 2020 – Apr. 2024

RESEARCH AND PROFESSIONAL EXPERIENCE

AI Research Assistant <i>PEG Digital Accelerator at University of Cincinnati</i>	Jun. 2026 – Present <i>Cincinnati, OH</i>
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- Developed an evidence-grounded enterprise GenAI prototype integrating multi-source retrieval, LLM reasoning, and a conversational interface over an internal knowledge base for R&D decision workflows.
- Designed repeatable evaluation across **five weighted dimensions**; analyzed grounding and reliability failures and presented findings to technical and business stakeholders.
- Integrated Chainlit, Azure AD, and Databricks AI Gateway with Claude Opus 4.7 while preserving source evidence and human review.

Lead Researcher, Climate Sustainability AI <i>ResearchDen</i>	Aug. 2026 – Present <i>Remote</i>
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- Direct a collaborative research team through problem formulation, literature review, methodology design, experimental planning, milestone coordination, and manuscript development.

Graduate Researcher and Simulation Technologist <i>Minnesota State University, Mankato</i>	Jan. 2025 – May 2026 <i>Mankato, MN</i>
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- Conducted federated-learning research on heterogeneous PlantVillage and PlantWild data using ResNet-18, EfficientViT, and ConvNeXt; compared centralized and federated performance over **30 training rounds**.
- Built reproducible analytics workflows and improved reporting efficiency by **25%** through structured data integration, validation, dashboards, and automated reporting.

AI/ML Intern; Machine Learning Intern <i>Fluentgrid Ltd.</i>	Jun. 2023 – Jun. 2024 <i>Visakhapatnam, India</i>
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- Developed computer-vision and analytics workflows for number-plate recognition, human-presence monitoring, and power-consumption analysis using Python, OCR, object detection, dashboards, and database-backed records.

SELECTED RESEARCH

Selective Agent Re-invocation via Evidence Sensitivity

- Designed a dependency-aware mechanism that re-executes only agents affected by changed evidence; evaluated quality retention, invocation savings, latency, and resource use across **13,500 legal, scientific, and medical instances**.
- Analyzed operating thresholds with bootstrap confidence intervals; observed an average **87.2% reduction in agent invocations** in pilot experiments while preserving downstream quality.

Auditable GraphRAG Multi-Agent Research Platform

- Built a Python platform combining semantic, keyword, and graph retrieval with specialist agents, Qdrant, persistent workspaces, REST APIs, tenant isolation, citations, and **100% source traceability**.
- Implemented regression coverage for retrieval, APIs, persistence, orchestration, and failure handling; completed **38/38 automated tests** successfully.

Privacy-Preserving and Compute-Efficient Language Models

- Evaluated LLaMA 3.3 70B, GPT-4o-mini, and Claude 3 Haiku in a federated framework with differential privacy, achieving an average **12.25% improvement** over single-model baselines across three datasets.
- Fine-tuned LLaMA-2 with **PEFT/LoRA and 4-bit quantization**, reducing compute requirements by **87%**; deployed an interactive Streamlit healthcare-information prototype.

PUBLICATIONS, SERVICE, AND RECOGNITION

Research Output: Six publications in applied AI, federated learning, computer vision, and data science; current manuscripts examine multi-agent reliability, evidence sensitivity, and causal auditing.

Scholarly Service: Peer reviewer for IEEE Transactions on Artificial Intelligence, Control and Cybernetics, and OzCHI.

Recognition: Graduate Student Research Grant; NVIDIA Deep Learning Institute credential.

TECHNICAL SKILLS

AI/ML: LLMs, AI agents, RAG, GraphRAG, NLP, deep learning, computer vision, PEFT/LoRA, quantization, federated learning, differential privacy, PyTorch, TensorFlow, scikit-learn

Research: experimental design, benchmarking, ablation studies, statistical analysis, bootstrap confidence intervals, model evaluation, failure analysis, technical writing

Systems and Data: Python, SQL, R, Spark, Kafka, FastAPI, REST APIs, Docker, Linux, Git, CI/CD, automated testing, Databricks, PostgreSQL, MongoDB, Qdrant, Power BI, Streamlit